

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Data structure

Practical – 3

Stack Application

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Class:SYIT	Batch:B1
Date of Assignment:2/7/2026	Date/Time of Submission: 2/7/2026

Stack Application:

Write a program to:

a. Implement a stack using an array.

Input

Overflow

Input

```
import array as arr

stack1 = arr.array('i')

si = int(input("Enter size of stack "))

print(stack1)

def Push():
    print("Push Operation")
    if(len(stack1)==si):
        print("Stack Overflow ")
    else:
        ele = int(input("Enter Element "))
        stack1.append(ele)
        print(stack1)
```

Output

```

Enter size of stack 2
array('i')
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit
Select Choice : 1
Push Operation
Enter Element 23
array('i', [23])
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit
Select Choice : 1
Push Operation
Enter Element 24
array('i', [23, 24])
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit
Select Choice : 1
Push Operation
Stack Overflow

```

Underflow

Input

```

import array as arr

stack1 = arr.array('i')

si = int(input("Enter size of stack "))

print(stack1)

def Push():
    print("Push Operation")
    if(len(stack1)==si):
        print("Stack Overflow ")
    else:
        ele = int(input("Enter Element "))
        stack1.append(ele)
        print(stack1)

def Pop():
    print("Pop Operation")
    if(isEmpty()==0):
        print("Stack Underflow")
    else:
        stack1.pop()
        print(stack1)

```

Output

```
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit  
Select Choice : 2  
Pop Operation  
array('i', [23])  
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit  
Select Choice : 2  
Pop Operation  
array('i')  
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit  
Select Choice : 2  
Pop Operation  
Stack Underflow
```

Input

practical3.py - C:/Users/itlab/OneDrive/priyanshi syit/practical3.py (3.14.6)

File Edit Format Run Options Window Help

```
import array as arr

stack1 = arr.array('i')

si = int(input("Enter size of stack "))

print(stack1)

def Push():
    print("Push Operation")
    if(len(stack1)==si):
        print("Stack Overflow ")
    else:
        ele = int(input("Enter Element "))
        stack1.append(ele)
        print(stack1)

def Pop():
    print("Pop Operation")
    if(isEmpty()==0):
        print("Stack Underflow")
    else:
        stack1.pop()
        print(stack1)

def Peek():
    print("Peek Operation")
    if(isEmpty()==0):
        print("Stack is Empty")
    else:
        print(stack1[-1])

def isEmpty():
    if len(stack1)==0:
        return 0

num = 0
while(num<5):
    print("1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit")
    num = int(input("Select Choice : "))

    if(num==1):
        Push()
```

```
num = 0
while (num<5):
    print("1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit")
    num = int(input("Select Choice : "))

    if (num==1):
        Push()
    elif (num==2):
        Pop()
    elif (num==3):
        Peek()
    elif (num==4):
        isEmpty()
    else:
        print("Program Closed")

print("Priyanshi Gupta")
```

Output

```
=====
Enter size of stack 1
array('i')
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit
Select Choice : 1
Push Operation
Enter Element 34
array('i', [34])
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit
Select Choice : 3
Peek Operation
34
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit
Select Choice : 1
Push Operation
Stack Overflow
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit
Select Choice : 2
Pop Operation
array('i')
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit
Select Choice : 2
Pop Operation
Stack Underflow
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit
Select Choice : 4
1. Push 2.Pop 3.Peek 4. isEmpty 5.Exit
Select Choice : 5
Program Closed
Priyanshi Gupta
```

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